

Problem Set #24

UMass Recreational Math Club

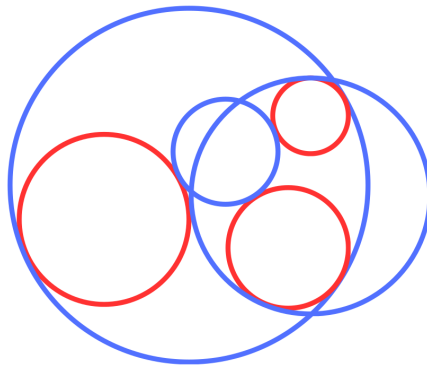
October 8th, 2025

Problem 1. What is the smallest three digit palindrome divisible by 18? Can you solve this without a brute force examination of possibilities?

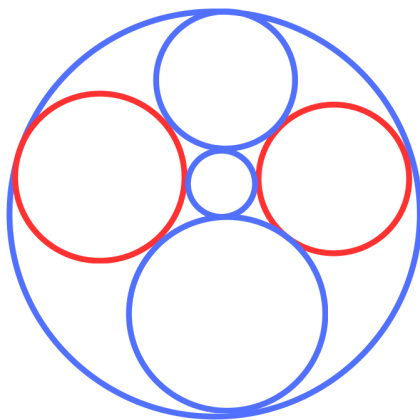
Problem 2. When Nia was learning about decimals, her teacher asked her to add two terminating decimals with a positive integer part. Then, Nia's teacher asked her to multiply those same two decimals. To Nia's surprise, the result of both computations was the same!

Both of the two numbers Nia's teacher gave her were non-whole positive numbers. When written as decimals without trailing zeroes, both of these numbers also had the same number of digits after the decimal point, and did not contain the digit 0. Find all possible pairs of numbers Nia's teacher could have given her.

Problem 3. Three red circles are placed in the plane. Your job is to place three more circles such that each of the six circles is tangent to exactly three others (one example is given by the blue circles below). In how many ways can you do this?



Problem 4 (Extra spicy). *Two* red circles are placed in the plane. Your job is to place *four* more circles such that each of the *six* circles is tangent to exactly *four* others (one example is given by the blue circles below). In how many ways can you do this?



Problem 5. For which parameters are these problems well-posed? That is, find all positive integers n, m, k such that given n planar circles in general position, there is a finite and possibly non-zero number of ways to place m circles such that each of the $n + m$ circles is tangent to exactly k others.